

ELECTRONIC ANSWER BOOK

I AGREE TO ABIDE BY THE SENATE POLICY ON CONDUCT OF ASSIGNMENTS, MID-TERM EXAM, AND FINAL EXAM

Instruction: Using Microsoft Word/WordPad/Notepad, type Student's name, Student ID number, type Chapter title and applicable Question Number for Assignment, Mid-Term Exam or Final Exam (as listed under Brightspace, Quizzes) before typing Solution below. One separate file for each Assignment/Mid-Term/Final Exam Question.

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Assignment – LVQ (3%) – Question 1:

Solution:

Input:

k	X_k	T_k
1	[0 0 1]	1
2	[0 1 1]	1
3	[1 1 0]	2
4	[1 0 0]	2

Initial Weights:

j	W_j	C_j
1	[0 1 1]	1
2	[1 1 0]	2

(a) Iteration 1: $g(1) = 0.5$

Presenting the input $X_1 = [0\ 0\ 1]$ of class T_1

$$\begin{aligned}d_1 &= (0 - 0)^2 + (0 - 1)^2 + (1 - 1)^2 \\&= 1.00\end{aligned}$$

$$\begin{aligned}d_2 &= (0 - 1)^2 + (0 - 1)^2 + (1 - 0)^2 \\&= 3.00\end{aligned}$$

d_1 is smaller, the weight vector W_1 of class C_1 will be updated.

The input X_1 and the weight W_1 are of the same class,

$$\begin{aligned}W_1 &= [0.00\ 1.00\ 1.00] + 0.5 \times ([0\ 0\ 1] - [0\ 1\ 1]) \\&= [0.00\ 0.50\ 1.00]\end{aligned}$$

Presenting the input $X_2 = [0 \ 1 \ 1]$ of class T_1

With updated weights $W_1 = [0.00 \ 0.50 \ 1.00]$ & $W_2 = [1.00 \ 1.00 \ 0.00]$

$$\begin{aligned} d_1 &= (0 - 0.00)^2 + (1 - 0.50)^2 + (1 - 1.00)^2 \\ &= 0.25 \end{aligned}$$

$$\begin{aligned} d_2 &= (0 - 1.00)^2 + (1 - 1.00)^2 + (1 - 0.00)^2 \\ &= 2.00 \end{aligned}$$

d_1 is smaller, the weight vector W_1 of class C_1 will be updated.

The input X_2 and the weight W_1 are of the same class,

$$\begin{aligned} W_1 &= [0.00 \ 0.50 \ 1.00] + 0.5 \times ([0 \ 1 \ 1] - [0.00 \ 0.50 \ 1.00]) \\ &= [0.00 \ 0.75 \ 1.00] \end{aligned}$$

Presenting the input $X_3 = [1 \ 1 \ 0]$ of class T_2

With updated weights $W_1 = [0.00 \ 0.75 \ 1.00]$ & $W_2 = [1.00 \ 1.00 \ 0.00]$

$$\begin{aligned} d_1 &= (1 - 0.00)^2 + (1 - 0.75)^2 + (0 - 1.00)^2 \\ &= 2.06 \end{aligned}$$

$$\begin{aligned} d_2 &= (1 - 1)^2 + (1 - 1)^2 + (0 - 0)^2 \\ &= 0.00 \end{aligned}$$

d_2 is smaller, the weight vector W_2 of class C_2 will be updated.

The input X_3 and the weight W_2 are of the same class,

$$\begin{aligned} W_2 &= [1.00 \ 1.00 \ 0.00] + 0.5 \times ([1 \ 1 \ 0] - [1 \ 1 \ 0]) \\ &= [1.00 \ 1.00 \ 0.00] \end{aligned}$$

Presenting the input $X_4 = [1 \ 0 \ 0]$ of class T_2

With updated weights $W_1 = [0.00 \ 0.75 \ 1.00]$ & $W_2 = [1.00 \ 1.00 \ 0.00]$

$$\begin{aligned} d_1 &= (1 - 0.00)^2 + (0 - 0.75)^2 + (0 - 1.00)^2 \\ &= 2.56 \end{aligned}$$

$$\begin{aligned} d_2 &= (1 - 1)^2 + (0 - 1)^2 + (0 - 0)^2 \\ &= 1.00 \end{aligned}$$

d_2 is smaller, the weight vector W_2 of class C_2 will be updated.

The input X_4 and the weight W_2 are of the same class,

$$\begin{aligned} W_2 &= [1.00 \ 1.00 \ 0.00] + 0.5 \times ([1 \ 0 \ 0] - [1 \ 1 \ 0]) \\ &= [1.00 \ 0.50 \ 0.00] \end{aligned}$$

After Iteration 1:

The weight vectors are

$$W_1 = [0.00 \ 0.75 \ 1.00]$$

$$W_2 = [1.00 \ 0.50 \ 0.00]$$

The sum of all the differences between elements obtained by subtracting W_2 from W_1 as $(W_1 - W_2)$ at the end of the first iteration (using $g(1) = 0.5$)

$$= (0.00 - 1.00) + (0.75 - 0.50) + (1.00 - 0.00)$$

$$= \mathbf{0.25}$$